

20260715: Initial commit, github repo set up, lit review

Done today

- I-9 Work authorization paperwork with HR
- HR orientation @ 9:00 AM
- Initial fork and unfork of gLike github repo
- Added `biblio` subdirectory
 - relevant `.pdf` files for papers and `.bibtex` file(s)
 - [\(REVIEW\) brandt2023promise.pdf](#)
 - Summary: Ancestral recombination graphs (ARGs) introduce a paradigm shift to human evolutionary genomics comparable to the coalescent. The primary improvement is inference of genealogical history rather than reducing genomic data to summary statistics.
 - Traditional methods infer genealogical history indirectly through summary statistics like the SFS, LD (linkage disequilibrium), haplotype length(s), F-statistics (like F_{ST}), or PCA (principal coordinate analyses).
 - Rather than utilizing n th-degree summary statistics, the ARG is n th-minus-one, i.e., one "level" closer to the actual biological process: Observed genome --> mutations --> underlying (true) ARG --> forces of demography, recombination and selection.
 - ARG advantages:
 - incorporation of recombination
 - increased statistical power; demographic events influence branch length, tree

- topology, recombination breakpoints / endpoints, local and global tree structure
 - ARG interpretability, e.g., analysis of N_e , migration, introgression, relatedness (and recency of relatedness), local / global ancestry, selection, mutation, age of mutation, genealogical neighbors / nearest neighbors
 - increased temporal resolution
- ARG challenges
 - exact inference is computationally impossible / nonfeasible (hence annealing strategy)
 - difficult to quantify uncertainty
 - large sample sizes
 - large parameter space
 - recombination hotspots (irregularity of rate of recombination across genome)
 - sequencing errors
- [fan2022genealogical.pdf](#)
 - introduces framework for measuring genetic relatedness using ARGs, rather than directly observing genetic variants
 - genetic variants are imperfect observations; however, by first estimating the ARG and then computing relatedness, we obtain a relationship matrix that is more robust to ascertainment bias, missing variants, and stochastic variation in mutations
 - eGRM (expected GRM) based on ARGs, averaging over all mutations
 - Infer ARG from genotype data
 - Relate or tsinfer+tsdate
 - Traverse *every* branch in *every* local tree
 - Assume mutations occur Poisson distributed across ARG tree branches
 - Weight each branch by length (generations), genomic interval, and mutation rate

- Compute expected contribution *of each branch* to pairwise relatedness
 - For each branch, identify descendants which inherit that branch
 - Add the branch's expected contribution to the relationship matrix
- Integrate over all possible mutations on the inferred genealogy to obtain the expectation of the conventional GRM conditional on the inferred ARG.
- Limitations
 - accuracy depends on the inferred ARG
 - Errors in ARG inference will propagate downstream
 - ARG inference is currently computationally demanding
- [fan2025likelihood.pdf](#)
 - new framework for inferring demographic history directly from genealogical trees as opposed to genetic variation data (SFS or LD)
 - given an ARG, compute likelihood for genealogies under coalescent theory and then infer piecewise-constant demographic histories
 - ?? Treat estimated genealogies as primary data, instead of sequence polymorphisms as primary (observed) data
 - *This is pretty weird to me.*
 - As mentioned in the reviews, observed mutations only provide indirect evidence of a genealogy, whereas the genealogy itself is the underlying process which generates genetic variation
 - For a genealogy with a known topology and branch length(s), the probability of observing the given sequence of coalescent events depends on the effective population size (N_e)

through time, i.e., $N_e(t)$

- Larger population --> longer waiting time between coalescent events and vice versa
- constant-size piece (epoch)-based demographic inference (like in dadi or fastsimcoal)
 - likelihood of model is derived from genealogy (based on coalescent time distribution) as opposed to allele frequencies
- composite likelihood, treating local genealogies as *approximately independent* observations
 - neighboring trees are correlated (subject to similar recombinatory forces)
- expectation-conditional maximization algorithm
 - Alternating between computing expected summary statistics under the current model and updating the effective population sizes for each epoch
 - relatively efficient and scales to genome-wide datasets
- ARG inference introduces uncertainty --> focus on estimated genealogies rather than requiring true, or assumed true, genealogies
- Limitations
 - Dependence on ARG accuracy (like `fan2022genealogical`)
 - composite likelihood approach relies on approximation of independent neighboring genealogies (which is not true due to recombination)
 - piecewise-constant size population models / epoch structure
 - Neutrality with respect to selection and migration / introgression (gene flow)
- [\(REVIEW\) lewanski2024era.pdf](#)
 - REVISIT this paper for: marginal trees, recombination nodes, ancestral haplotypes, ARG terminology, local / global genealogy

(concepts)

- REVISIT this paper for: conceptual link between ARG and coalescent theory / coalescent trees
 - Without recombination, a single genealogy can explain the entire chromosome
 - ARGs allow us to incorporate recombination, by which different genomic regions possess different genealogies -- ARGs connect these genealogies
- REVISIT this paper for reference on ARG inference software: ARGweaver, Relate, tsinfer, tsdate, GEVA, RENT+, Tree sequences
 - @jonmah add in citations for these if relevant to our project
- Tree sequences / tree sequence data structure
 - Adjacent genomic regions usually share nearly identical genealogies (similarity of recombination rate, recombination breakpoints)
 - Tree sequences allow us to store the *shared structure* rather than storing the trees individually / independently, reducing need for memory and computation
- Similar to brand2023promise, this review mentions demographic inference, mutation dating, gene flow (migration / introgression), as well as medical genetic applications (genotype imputation, fine mapping, rare variant id / interpretation, IBD inference)
- [link2023tree.pdf](#)
 - sycamore: genealogy-based approach to quantitative trait locus (QTL) mapping.
 - instead of conventional SNP-based genetic relationship matrices, Link et al. uses expected genetic relationship matrices (eGRMs), similar to fan2022genealogical

- GWAS / QTL lens, is this improved by testing local genealogical relatedness as opposed to individual SNP genotypes?
- linear mixed model (LMM) where the random-effect covariance matrix is given by the local eGRM as opposed to something from SNP genotype data
 - For each genomic window:
 - compute the local eGRM
 - fit a LMM
 - test whether the local genealogy explains phenotypical variance
 - This yields a tree-based (ancestry-based) association scan across the genome
- Application to human genomic datasets, comparing tree-based mapping to SNP-based mapping
- Primary advantages for tree-based QTL-mapping are improved statistical power for untyped, poorly imputed, or weakly tagged causal variants
- Limitations, as per previous / other papers: dependence on ARG quality, computational cost of ARG inference (sample size), assumption of genetic additive effects
- [\(REVIEW\) nielsen2025inference.pdf](#)
 - REVISIT for more concrete / quantitative / statistical breakdown of ARGs (thank you Rasmus)
 - REVISIT for inference algo, statistical models, computational complexity, limitations of methods
 - Core challenge of ARGs is super-exponential growth of parameter space with respect to sample size and recombination events, yielding a computationally nonfeasible likelihood calculation, motivating approximate algorithms
 - Strategies include:

- MCMC (markov chain monte carlo), such as ARGweaver
 - probabilistic and allows for estimation of uncertainty, but slow and scales poorly
- Sequential approximation (sequentially building local trees across chromosomes / the genome), such as Relate and tsinfer
 - much faster but doesn't allow for simultaneous fitting, as computation is done sequentially
- HMM (hidden markov model)
- Tree sequence methods (similar to lewanski2024era)
- Similar to other reviews, mentions applicaiton to demographic inference, selection inference, introgression, medical (disease) genetics, and mutation age estimation
- [tang2025genealogy.pdf](#)
 - ARG-based framework for studying population structure of admixed populations.
 - ARGs used to characterize how different ancestral populations contribute to present-day genomes
 - Essentially extension of ARG-based demographic inference to include ancestry (multiple ancestral populations)
 - Traditional analyses consists of inferring local ancestry, and then analyzing ancestry tracts using allele frequencies or PCA
 - ... Why don't we just analyze the ARGs?
 - Genealogies (ARGs) provide a richer / more detailed representation of ancestry than local ancestry tracts, which fail to capture differences between similar ancestry groups that have distinct demographic histories
 - Method
 - Infer ARG from sampled genomes
 - From local genealogies across the

genome, compute genetic relatedness for different ancestral backgrounds

- How do lineages cluster within genealogical trees across genomic regions?
- Summarize genealogical relationships to identify ancestry-specific patterns of population structure, i.e., estimation of ancestry-specific genetic relationships
 - for each ancestral background, we compute how closely related individuals are within the corresponding part of the local genealogy
 - because recombination yields different genealogies across the genome, the approach integrates information from many many local trees across the genome
 - distinguish between recent vs. shared ancestry, different subgroups within the "same" ancestry, and heterogenous demographic histories within ancestral populations

■ Advantages

- higher resolution inference of ancestry-specific structure than allele frequency methods
- better characterization of admixed populations (evolutionary relationships amongst ancestry tracts)

■ Limitations

- dependent on accuracy of ARG
- ARG inference is computationally expensive
- sequence errors or errors in local genealogies can reduce power

■ [zhu2026leveraging.pdf](#)

- similar to sycamore, ARG-based method for

GWAS and mixed-model analysis of complex traits

- SNP-based genetic relationship matrix (GRM) with an *ARG-derived relationship matrix* that captures shared ancestry directly
- instead of "how many alleles do two individuals share" (traditional genetic relationship matrix), we ask "how much evolutionary history do two individuals share" (ARG-based relationship matrix)
 - ideally yielding a scenario closer to the actual underlying biological process
- ARG-derived relationship matrix method:
 - for each genomic region,
 - infer local genealogical trees from an ARG
 - measure shared ancestry between individuals
 - aggregate across the genome (integrate over genealogical branches)
- sped up using tree-sequence data structures and fast matrix-vector operations
- Limitations
 - dependent on ARG quality
 - computational complexity
 - model assumptions of additive genetic effects, "accurate" phenotype mapping, and a normal distribution of random effects
- Added `lab_notebook` subdirectory
 - Lab notebook entries for progress and TODO
 - Weekly entries:
 - [20260715.md: Initial commit, github repo set up, introduction brainstorming](#)

Review of literature review

My key takeaway is that literature is pointing towards a shift away from using observations of polymorphism data as input (SNP-based methods) and towards using genealogical relationships across the genome as input (ARGs). This is limited by the need for accurate ARGs (similar to how other methods rely on accurate SNP data), and the need for computational complexity (due to the superscaling nature of ARG parameter space contingent on sample size and recombination events).

This clearly outlines the motivation of the project, being to optimize ARG inference out of / away from local minima while also improving upon computational inefficiencies.

Github repo set up

- [Github repository is here](#). I forked it off of [Caoqi's repository for gLike](#) and then unforked it in case we want to change visibility to private during the project. Note that it cannot be re-forked, but that's not particularly important so long as we correctly attribute the source code (which is the plan).
- The way I plan to set up the repository is to also use it as a lab notebook. Specifically, I have added a `biblio` subdirectory (for storing papers and an eventual `.bibtex` file) and a `lab_notebook` subdirectory, for `.md` documents like this one to help keep track of the project.
- The way I have set up projects in the past is to have the "manuscript" github repository be a copy or fork/unfork of the "lab notebook" github repository, thus if anyone wanted to access the full history of the project, they could. Here is an example of a [publication github repository that I've set up in the past](#).

TODO

My goal for tomorrow is to start digging through some of the code (installation and messing around).